# BIG SPRING, TX, USA

WMO: **720271**

Lat: **32.213N** Lon: **101.521W** Elev: **784** StdP: **92.25** Time Zone: **-6.00 (NAC)** Period: **04-19** WBAN: **03044**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **-6.2** | **-4.1** | **-17.0** | **0.9** | **2.6** | **-14.3** | **1.2** | **4.5** | **14.1** | **11.8** | **12.2** | **11.5** | **3.2** | **20** | **0.550** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **12.2** | **38.1** | **19.5** | **37.5** | **19.5** | **36.4** | **20.1** | **23.0** | **30.9** | **22.5** | **30.3** | **21.9** | **29.7** | **5.0** | **150** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **21.3** | **17.6** | **24.9** | **20.8** | **17.1** | **24.6** | **20.1** | **16.3** | **24.6** | **72.2** | **30.7** | **70.1** | **30.1** | **68.2** | **29.7** | **32.5** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **11.9** | **10.8** | **9.3** | DB | **-9.3** | **41.4** | **2.6** | **3.2** | **-11.2** | **43.7** | **-12.8** | **45.6** | **-14.2** | **47.4** | **-16.2** | **49.7** | (4) |
| (5) | | | | WB | **-10.6** | **25.4** | **2.3** | **2.6** | **-12.3** | **27.3** | **-13.7** | **28.8** | **-15.0** | **30.2** | **-16.7** | **32.1** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | | DBAvg | **18.5** | **7.4** | **9.8** | **14.6** | **19.0** | **23.1** | **27.9** | **28.5** | **28.0** | **24.1** | **18.8** | **12.7** | **7.9** | (6) |
| (7) | | | DBStd | **8.80** | **4.99** | **5.80** | **5.28** | **4.86** | **4.59** | **2.98** | **2.56** | **2.67** | **3.43** | **4.89** | **4.99** | **5.05** | (7) |
| (8) | | | HDD10.0 | **336** | **110** | **68** | **19** | **3** | **0** | **0** | **0** | **0** | **0** | **4** | **31** | **101** | (8) |
| (9) | | | HDD18.3 | **1334** | **338** | **240** | **136** | **50** | **13** | **0** | **0** | **0** | **2** | **53** | **177** | **324** | (9) |
| (10) | | | CDD10.0 | **3451** | **30** | **63** | **164** | **273** | **406** | **538** | **573** | **558** | **424** | **277** | **110** | **36** | (10) |
| (11) | | | CDD18.3 | **1407** | **0** | **2** | **22** | **69** | **160** | **288** | **314** | **299** | **177** | **68** | **7** | **1** | (11) |
| (12) | | | CDH23.3 | **16671** | **22** | **84** | **328** | **873** | **1895** | **3520** | **3873** | **3512** | **1765** | **666** | **111** | **23** | (12) |
| (13) | | | CDH26.7 | **8402** | **1** | **16** | **93** | **358** | **958** | **1944** | **2127** | **1882** | **776** | **227** | **18** | **1** | (13) |
| (14) | Wind | | WSAvg | **4.5** | **4.2** | **4.6** | **5.2** | **5.6** | **5.3** | **5.3** | **4.3** | **3.9** | **3.9** | **4.1** | **4.1** | **4.1** | (14) |
| (15) | Precipitation | | PrecAvg | **500** | **18** | **24** | **32** | **37** | **61** | **66** | **50** | **62** | **57** | **43** | **29** | **18** | (15) |
| (16) | | | PrecMax | **878** | **50** | **73** | **97** | **114** | **181** | **132** | **145** | **134** | **151** | **109** | **178** | **99** | (16) |
| (17) | | | PrecMin | **262** | **0** | **0** | **4** | **2** | **7** | **6** | **4** | **4** | **3** | **0** | **0** | **0** | (17) |
| (18) | | | PrecStd | **151** | **15** | **22** | **27** | **31** | **45** | **40** | **40** | **37** | **41** | **33** | **40** | **23** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **26.2** | **28.8** | **32.4** | **36.2** | **38.9** | **41.1** | **39.0** | **40.0** | **37.5** | **34.2** | **28.7** | **26.3** | (19) |
| (20) | | | MCWB | | **12.1** | **12.2** | **14.0** | **15.5** | **17.0** | **19.0** | **20.0** | **19.7** | **19.3** | **18.4** | **14.8** | **11.8** | (20) |
| (21) | | 2% | DB | | **22.8** | **26.1** | **29.2** | **33.7** | **37.2** | **38.3** | **37.8** | **37.8** | **35.1** | **31.8** | **26.4** | **22.9** | (21) |
| (22) | | | MCWB | | **10.5** | **11.9** | **13.7** | **15.3** | **17.2** | **19.7** | **20.2** | **20.1** | **19.5** | **18.4** | **13.9** | **11.6** | (22) |
| (23) | | 5% | DB | | **19.2** | **23.2** | **27.1** | **30.9** | **34.7** | **37.3** | **37.1** | **36.7** | **33.4** | **29.1** | **24.0** | **18.9** | (23) |
| (24) | | | MCWB | | **9.2** | **11.4** | **13.2** | **14.6** | **17.3** | **19.7** | **20.5** | **20.4** | **19.7** | **17.2** | **13.8** | **10.1** | (24) |
| (25) | | 10% | DB | | **16.3** | **20.1** | **24.9** | **28.0** | **32.3** | **35.7** | **35.7** | **34.9** | **31.2** | **27.2** | **21.8** | **16.3** | (25) |
| (26) | | | MCWB | | **8.1** | **10.1** | **13.2** | **14.5** | **17.5** | **20.2** | **20.7** | **20.3** | **19.2** | **16.9** | **12.5** | **8.8** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **15.1** | **15.9** | **18.8** | **20.0** | **22.0** | **23.6** | **23.6** | **23.5** | **23.5** | **21.9** | **18.0** | **15.8** | (27) |
| (28) | | | MCDB | | **18.9** | **21.2** | **24.4** | **26.2** | **29.0** | **31.9** | **31.7** | **31.1** | **29.7** | **27.1** | **23.3** | **19.3** | (28) |
| (29) | | 2% | WB | | **12.2** | **14.3** | **17.3** | **19.0** | **21.3** | **22.8** | **22.8** | **22.8** | **22.3** | **20.9** | **16.7** | **13.5** | (29) |
| (30) | | | MCDB | | **18.2** | **20.4** | **23.6** | **25.0** | **28.0** | **31.2** | **31.2** | **30.7** | **28.0** | **26.3** | **21.7** | **18.1** | (30) |
| (31) | | 5% | WB | | **10.6** | **12.7** | **16.0** | **17.9** | **20.6** | **22.1** | **22.2** | **22.2** | **21.6** | **20.0** | **15.5** | **11.7** | (31) |
| (32) | | | MCDB | | **18.4** | **20.3** | **22.1** | **24.3** | **27.2** | **30.4** | **30.9** | **29.9** | **27.4** | **25.5** | **20.3** | **18.1** | (32) |
| (33) | | 10% | WB | | **8.9** | **11.1** | **14.7** | **16.9** | **19.9** | **21.6** | **21.7** | **21.7** | **20.9** | **19.1** | **14.1** | **10.0** | (33) |
| (34) | | | MCDB | | **16.2** | **19.3** | **21.1** | **24.0** | **26.9** | **29.9** | **30.5** | **29.3** | **27.2** | **24.8** | **19.8** | **15.9** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **14.2** | **14.6** | **14.8** | **15.1** | **13.8** | **13.1** | **12.2** | **12.1** | **11.9** | **13.3** | **13.8** | **13.2** | (35) |
| (36) | | 5% DB | MCDBR | | **19.2** | **19.4** | **17.8** | **18.1** | **17.2** | **15.3** | **13.9** | **14.0** | **14.7** | **15.6** | **16.9** | **18.1** | (36) |
| (37) | | | MCWBR | | **9.6** | **9.5** | **7.9** | **7.3** | **6.3** | **4.8** | **4.4** | **4.3** | **5.0** | **6.1** | **8.0** | **9.4** | (37) |
| (38) | | 5% WB | MCDBR | | **16.6** | **16.5** | **14.0** | **14.0** | **13.7** | **12.9** | **11.8** | **11.7** | **10.7** | **11.9** | **12.4** | **15.0** | (38) |
| (39) | | | MCWBR | | **9.2** | **9.3** | **7.4** | **7.3** | **6.1** | **4.5** | **4.1** | **4.1** | **4.2** | **6.0** | **7.3** | **8.9** | (39) |
| (40) | Clear-Sky Solar Irradiance | | taub | | **0.269** | **0.280** | **0.304** | **0.340** | **0.356** | **0.385** | **0.398** | **0.377** | **0.359** | **0.323** | **0.293** | **0.275** | (40) |
| (41) | | | taud | | **2.579** | **2.554** | **2.498** | **2.391** | **2.417** | **2.375** | **2.395** | **2.424** | **2.429** | **2.516** | **2.534** | **2.563** | (41) |
| (42) | | | Ebn at Noon | | **970** | **987** | **980** | **948** | **930** | **898** | **884** | **901** | **907** | **923** | **930** | **943** | (42) |
| (43) | | | Edh at Noon | | **79** | **90** | **103** | **119** | **118** | **122** | **119** | **114** | **109** | **93** | **82** | **77** | (43) |
| (44) | All-Sky Solar Radiation | | RadAvg | | **3.39** | **4.17** | **5.41** | **6.65** | **7.00** | **7.49** | **7.13** | **6.61** | **5.46** | **4.45** | **3.55** | **3.00** | (44) |
| (45) | | | RadStd | | **0.34** | **0.48** | **0.40** | **0.42** | **0.57** | **0.38** | **0.42** | **0.31** | **0.39** | **0.55** | **0.39** | **0.29** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (17 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **+112** | **+78** | (47) |

Nomenclature: See separate page